# Sathwik Munigala

B.Tech — Mechanical Engineering

**IIITDM KURNOOL, Batch: 2023–2027**

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## CAREER OBJECTIVE

Final-year B.Tech student in Mechanical Engineering with a minor in Robotics, with hands-on experience across the complete robot development cycle: mechanical design, electronics, simulation, reinforcement learning and deployment on real hardware. Seeking a full-time robotics role in legged and humanoid robots, robot learning, simulation, or autonomous navigation and manipulation with ROS 2.

## EDUCATIONAL QUALIFICATIONS

| Degree / Qualification | Institution | Year | Score / CGPA |
|---|---|---|---|
| B.Tech in Mechanical Engineering and Minor Robotics | Indian Institute of Information Technology Design & Manufacturing, Kurnool | 2023–2027 | 7 / 10 |
| Intermediate (MPC) | Narayana Junior College, Madhapur, Hyderabad | 2021–2023 | 95.5% |
| Class X | Narayana High School, Tarnaka, Hyderabad | 2021 | 10 / 10 |

## TECHNICAL SKILLS & KNOWLEDGE

| | |
|---|---|
| **Robotics and Simulation** | NVIDIA Isaac Sim, Isaac Lab, Gazebo, ROS 2 Humble, micro-ROS, Nav2, slam_toolbox, AMCL, MoveIt 2, ros2_control, RViz, URDF and USD robot modelling |
| **Robot Learning and Control** | Reinforcement learning (PPO), reward design, curriculum learning, forward and inverse kinematics, Denavit-Hartenberg parameters, gait and foot trajectory design, sim-to-real transfer |
| **Perception** | Sensor fusion (LiDAR, ultrasonic, depth camera, IMU), OpenCV, CNN-based shape recognition |
| **Programming** | Python, PyTorch, Embedded C, Linux (Ubuntu), Git |
| **Embedded and Electronics** | Raspberry Pi, ESP32, Arduino, PCA9685 PWM servo driver, I2C, servo calibration, power distribution and wiring |
| **Design Software** | SolidWorks, Fusion 360, Onshape, Siemens NX, AutoCAD, MATLAB |
| **Manufacturing** | CNC Programming, Wire EDM, Welding Processes, Milling Machine, Turning, Metrology & QC, 3D Printing |
| **Mechanical Design and Analysis** | Machine design, mechanism design, kinematic analysis, gear design, tolerance analysis, GD&T, Design for Manufacturing (DFM), IS codes and BIS Standards |
| **Domain Subjects** | Kinematics of Machinery, Dynamics of Machinery, Machine Design, Heat Transfer |

## ACADEMIC PROJECTS

**Quadruped Robot: Design, Learning-Based Walking and Sim-to-Real** — Final Year Project 2026

Developed a 12-degree-of-freedom four-legged robot through the complete engineering cycle: mechanical design, robot modelling, electronics, simulation, control and transfer to the real hardware. Walking was developed in two ways. The first is learning-based, where a locomotion policy is trained with reinforcement learning (PPO) in NVIDIA Isaac Sim and Isaac Lab. The second is model-based, where a trot gait is generated from the kinematics of the legs and a learned policy keeps the robot balanced. The robot responds to forward, backward, sideways and turning commands, and the motion produced in simulation is reproduced on the real robot in real time. In future, the robot will be extended to walk on rough terrain and climb stairs. Demonstration videos of the simulation-to-real transfer, showing the real robot and the simulation side by side, and further technical details are available on request.

**Design and Fabrication of Manual Gear Transmission System** — 3rd Year 2025–2026

Executed the complete end-to-end development of a manual transmission system. Performed kinematic analysis and 3D modelling in SolidWorks, followed by fabrication and assembly of the gears. Focused on gear ratio optimisation, synchronizer mechanism design, and testing structural integrity.

## INTERNSHIP / INDUSTRIAL TRAINING

**CAD Design Intern — Primeform Labs LLP** May 2026 – Jun 2026, Hyderabad

- Executed the end-to-end product design cycle for industrial pressure valves and custom connectors, transitioning from

initial rough conceptual sketches to final production-ready assemblies.
- Conducted analytical and mechanical calculations to determine optimal dimensions, wall thicknesses, and material safety factors meeting industry manufacturing standards.
- Developed high-fidelity 3D CAD models, robust parametric designs, and production-ready 2D CAD drawing files using strict DFM (Design for Manufacturing) guidelines to ensure flawless machine shop execution.

## PERSONAL PROJECTS

**Bipedal Robot: Kinematics and RL Locomotion (ROS 2, Isaac Sim, Isaac Lab)** — Ongoing 2026

- Designed a small humanoid-style biped in Onshape (hip, knee and ankle joints, arms, real joint limits), exported it to URDF and built a ROS 2 package to view and move it joint by joint in RViz.
- Derived the DH parameters and the forward and inverse kinematics of the legs and arms as a base for trajectory planning.
- Set up the robot in Isaac Sim with foot contact sensors and a body IMU, validated against the simulator's own values.
- Trained a walking policy with PPO in a parallel Isaac Lab environment; it balances, follows speed and turning commands and keeps a regular left–right stepping rhythm on flat ground.

**Autonomous Mobile Robot in Isaac Sim (ROS 2, Nav2)** — GitHub Nov 2026

- Designed a differential-drive robot in SolidWorks and exported it to URDF and USD, with a 360° LiDAR, four ultrasonic sensors, a depth camera and a 9-axis IMU simulated in NVIDIA Isaac Sim 5.1.
- Wrote a sensor-fusion node that merges the LiDAR and ultrasonic scans into one scan, so the costmaps also see low obstacles below the LiDAR plane.
- Built three navigation modes: Nav2 navigation with AMCL on a saved map, navigating while mapping an unknown maze with slam_toolbox, and a back-and-forth corridor lane-coverage planner that reaches about 99% floor coverage and writes a coverage report. Added a GUI to launch and stop each task.

**Dual-Arm Pattern Drawing and Shape Recognition (ROS 2, MoveIt 2, Gazebo)** — GitHub Oct 2026

- Set up a Franka Panda and a UR5 in one Ignition Gazebo world, each with its own namespace, controller manager and MoveIt 2 move group.
- The Panda draws shapes (circle, polygon, star, spiral and more) on a table with a pen end effector; the UR5 scans the drawing with a wrist camera.
- A small CNN classifies the drawn shape, cross-checked by an OpenCV corner count, and a report compares the drawn ink against the planned path.

**Quadruped Robot Navigation (Research Project)**

- Developed a simulated quadruped robot for autonomous navigation using the ROS2 Nav2 framework. Implemented localization and multi-planner navigation pipelines and conducted benchmarking analysis as part of an ongoing research article.
- **Extension:** migrated the pipeline to the latest NVIDIA Isaac Sim release and built a fully automated benchmarking loop that launches Isaac Sim, runs a planner end-to-end, then tears it down and relaunches with the next planner — sequencing all planner runs unattended with no manual intervention.

**Motor Control via ROS2 and Micro-ROS (ESP32)**

- Real-time DC motor control using Micro-ROS on embedded systems (ESP32/Arduino) with ROS 2 integration.

**ANYmal-C Quadruped Locomotion Training (PPO + Isaac Gym)**

- Trained the ANYmal-C quadruped robot to walk using Proximal Policy Optimisation (PPO) in the NVIDIA Isaac Gym physics simulator. Implemented actor-critic neural networks for robust locomotion control across various terrains.

## ACHIEVEMENTS & CERTIFICATIONS

- Under publication — Technical Paper: "Comparative Analysis of Heterogeneous Planner Stack Structure for Robust Quadruped Navigation"

## CO-CURRICULAR & EXTRA-CURRICULAR

- Robotics Club Head (Coordinator), 2024–2026
- Core Member — Organised technical workshops and mechanical fabrication sessions; mentored students in mechanical design and robotics system development.
- NSS Volunteer — Participated in blood donation, literacy drives, and disaster relief.